A Global Bang–Bang Value Formula for the Bounded Double Integrator

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Abstract

For the physical double integrator $\dot{x} = v$, $\dot{v} = u$, $|u| \le a$, we close the minimum-time problem to rest at the origin on the entire state plane. The switching function $F_a = x + v|v|/(2a)$ selects either direct braking or a unique one-switch control. We give the two arc lengths, switch state and a closed value function, and verify exact termination and the Hamilton–Jacobi equation off the nonsmooth switching curve. A sharp rearrangement bound for the endpoint control moments supplies a global sufficiency proof; the affine Pontryagin switching function is an independent structural check. Origin, zero acceleration, reflection and parabolic scaling are retained explicitly. Executable exact/high-precision, symbolic, replay and hostile-mutation audits accompany the theorem.

1 Convention and synthesis

Fix $a > 0$ and let $T_a(x, v)$ be the least physical time for

$$\dot{x} = v, \qquad \dot{v} = u, \qquad |u| \le a$$

to reach $(0, 0)$. Set

$$F_a(x, v) = x + \frac{v|v|}{2a}.$$

This convention fixes the state ordering, acceleration sign, terminal state and time clock; no rescaled or fitted clock is used.

Theorem 1 (Whole-plane minimum-time synthesis)**.** *At $(0, 0)$, $T_a = 0$. If $F_a = 0$ and $v \ne 0$, the direct brake $u = -a \operatorname{sign} v$ is optimal and $T_a = |v|/a$.*

If $F_a \ne 0$, let $s = \operatorname{sign} F_a$ and

$$D = \frac{v^2}{2a^2} + \frac{sx}{a}.$$

Then $D > 0$ and

$$t_1 = \frac{sv}{a} + \sqrt{D} \ge 0, \qquad t_2 = \sqrt{D} > 0, \qquad T_a(x, v) = \frac{sv}{a} + 2\sqrt{D}. \tag{1}$$

Up to null sets, the optimal control is $u = -sa$ for time t_1 and $u = sa$ for time t_2. Its switch state is

$$(x_1, v_1) = \left(\frac{saD}{2}, -sa\sqrt{D}\right), \tag{2}$$

and the second arc terminates at $(0, 0)$. Moreover,

$$T_a(-x, -v) = T_a(x, v), \qquad T_a(\lambda^2 x, \lambda v) = \lambda T_a(x, v) \quad (\lambda > 0). \tag{3}$$

Algebraic construction. The assertion $D > 0$ follows directly from $sF_a > 0$; if $sv < 0$, that same strict inequality gives $\sqrt{D} > |v|/a$, and otherwise $t_1 \ge 0$ is immediate. Integrating the first constant-control arc gives (2). Since $v_1|v_1| = -sa^2 D$, (2) lies on $F_a = 0$. The second arc has duration $\sqrt{D}$, so $v_1 + sa\sqrt{D} = 0$ and $x_1 + v_1\sqrt{D} + saD/2 = 0$. Reflection and scaling follow by substitution in (1). The global lower bound is proved below. ☐

2 Two independent optimality certificates

For a normal minimum-time extremal, the costate satisfies $\dot{p}_x = 0$ and $\dot{p}_v = -p_x$. Hence p_v is affine in time and the minimizing control $u = -a \operatorname{sign} p_v$ switches at most once. This explains the structure but, as a necessary condition, is not by itself the global sufficiency proof.

Off $F_a = 0$, differentiating (1) gives

$$(T_a)_x = \frac{s}{a\sqrt{D}}, \qquad (T_a)_v = \frac{s}{a} + \frac{v}{a^2\sqrt{D}}.$$

The inequality used above gives $\operatorname{sign}(T_a)_v = s$, and therefore

$$1 + v(T_a)_x - a|(T_a)_v| = 0. \tag{4}$$

Thus the displayed control attains the minimum in the HJB equation wherever the value is smooth.

3 Sharp reachable-moment lower bound

The missing sufficiency step has a particularly short source-local proof. If a control reaches rest in time T, integration of velocity and position gives

$$\int_0^T u(t)\,dt = -v, \qquad \int_0^T t\,u(t)\,dt = x. \tag{5}$$

Lemma 1 (Sharp bounded-control moment interval)**.** *Every measurable u satisfying $|u| \le a$ and the first identity in (5) obeys*

$$-\frac{aT^2}{4} - \frac{vT}{2} + \frac{v^2}{4a} \le \int_0^T tu(t)\,dt \le \frac{aT^2}{4} - \frac{vT}{2} - \frac{v^2}{4a}. \tag{6}$$

The upper endpoint is attained by placing $-a$ first and $+a$ last; the lower endpoint reverses that order. Apart from null sets these are the only endpoint controls.

Proof. Write $u = -a + 2aw$ with $0 \le w \le 1$. Its prescribed integral fixes $\int_0^T w = (T - v/a)/2$. The bathtub principle maximizes $\int_0^T tw(t)\,dt$ by putting $w = 1$ on the latest interval of that measure, and minimizes it by putting $w = 1$ on the earliest interval. Direct integration gives (6), including the equality cases. □

For $F_a > 0$, the least feasible time makes the upper inequality in (6) an equality; solving it gives (1) with $s = 1$. For $F_a < 0$, the lower inequality is active and gives (1) with $s = -1$. The equality cases are precisely the constructed controls. Consequently no shorter transfer is admissible, which completes the global proof of Theorem 1. In viscosity language, the continuous value solves the HJB equation globally, with (4) its classical restriction away from the switch.

4 Boundary atlas

On $F_a = 0$, both adjacent formulas meet $|v|/a$, but the value need not be classically differentiable. The origin is the zero-duration case. At $a = 0$, velocity is constant; only $(0, 0)$ can reach rest at the origin in finite time, so every other value is infinite. These branches are not obtained by blindly dividing a positive-a formula by zero.

Source note

Romano and Curti [1] treat minimum-time bounded normal LTI systems, reduction to an origin transfer, and the double-integrator context. We claim no priority for bang–bang synthesis or Pontryagin's principle; the release contribution is the convention-locked theorem/boundary synthesis and reproducible audit.

References

[1] M. Romano and F. Curti, *Time-optimal control of linear time invariant systems between two arbitrary states*, *Automatica* 120 (2020), 109151. DOI: 10.1016/j.automatica.2020.109151.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, target-zero or operator claim. **Data and code.** The theorem, canonical ledger and independent audits are released with HCS-C222. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.